

From Time Dilation to Mass Variation:

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Abstract

This updated work presents the essential mathematical formulations of time-mass duality theory, building upon the comprehensive geometric foundations established in the field-theoretic derivation of the β parameter. The theory establishes a duality between two complementary descriptions of reality: the standard view with time dilation and constant rest mass, and the T0 model with absolute time and variable mass. Central to this framework is the intrinsic time field $T(x, t) = \frac{1}{\max(m, \omega)}$ (in natural units where $\hbar = c = \alpha_{\text{EM}} = \beta_{\text{T}} = 1$), which enables a unified treatment of massive particles and photons through the three fundamental field geometries: localized spherical, localized non-spherical, and infinite homogeneous. The mathematical formulations include complete Lagrangian densities with strict dimensional consistency, incorporating the derived parameters $\beta = 2Gm/r$, $\xi = 2\sqrt{G} \cdot m$, and the cosmic screening factor $\xi_{\text{eff}} = \xi/2$ for infinite fields. All equations maintain perfect dimensional consistency and contain no adjustable parameters.

1 Introduction: Updated T0 Model Foundations

This updated mathematical formulation builds upon the comprehensive field-theoretic foundation established in the T0 model reference framework. The time-mass duality theory now incorporates the complete geometric derivations and natural units system that demonstrate the fundamental unity of quantum and gravitational phenomena.

Fundamental Postulate: Intrinsic Time Field

The T0 model is based on the fundamental relationship between time and mass expressed through the intrinsic time field:

$$T(x, t) = \frac{1}{\max(m(x, t), \omega)} \tag{1}$$

✓ **Dimensional verification:** $[T(x, t)] = [1/E] = [E^{-1}]$ in natural units

This field satisfies the fundamental field equation derived from geometric principles:

$$\nabla^2 m(x, t) = 4\pi G \rho(x, t) \cdot m(x, t) \quad (2)$$

Dimensional verification: $[\nabla^2 m] = [E^2][E] = [E^3]$ and $[4\pi G \rho m] = [1][E^{-2}][E^4][E] = [E^3]$ ✓

Three Fundamental Field Geometries

The complete T0 framework recognizes three distinct field geometries with specific parameter modifications:

T0 Model Parameter Framework

Localized Spherical Fields:

$$\beta = \frac{2Gm}{r} \quad [1] \quad (3)$$

$$\xi = 2\sqrt{G} \cdot m \quad [1] \quad (4)$$

$$T(r) = \frac{1}{m_0}(1 - \beta) \quad (5)$$

Localized Non-spherical Fields:

$$\beta_{ij} = \frac{r_{0ij}}{r} \quad (\text{tensor}) \quad (6)$$

$$\xi_{ij} = 2\sqrt{G} \cdot I_{ij} \quad (\text{inertia tensor}) \quad (7)$$

Infinite Homogeneous Fields:

$$\nabla^2 m = 4\pi G \rho_0 m + \Lambda_T m \quad (8)$$

$$\xi_{\text{eff}} = \sqrt{G} \cdot m = \frac{\xi}{2} \quad (\text{cosmic screening}) \quad (9)$$

$$\Lambda_T = -4\pi G \rho_0 \quad (10)$$

Practical Simplification Note

For practical applications: Since all measurements in our finite, observable universe are performed locally, only the **localized spherical field geometry** (first case above) is required:

$\xi = 2\sqrt{G} \cdot m$ and $\beta = \frac{2Gm}{r}$ for all applications.

The other geometries are shown for theoretical completeness but are not needed for experimental predictions.

Natural Units Framework Integration

The complete natural units system where $\hbar = c = \alpha_{\text{EM}} = \beta_{\text{T}} = 1$ provides:

- Universal energy dimensions: All quantities expressed as powers of $[E]$
- Unified coupling constants: $\alpha_{\text{EM}} = \beta_{\text{T}} = 1$ through Higgs physics
- Connection to Planck scale: $\ell_{\text{P}} = \sqrt{G}$ and $\xi = r_0/\ell_{\text{P}}$
- Fixed parameter relationships: No adjustable constants in the theory

.2 Complete Field Equation Framework

Spherically Symmetric Solutions

For a point mass source $\rho = m\delta^3(\vec{r})$, the complete geometric solution is:

$$m(x, t)(r) = m_0 \left(1 + \frac{2Gm}{r} \right) = m_0(1 + \beta) \quad (11)$$

Therefore:

$$T(r) = \frac{1}{m(x, t)(r)} = \frac{1}{m_0} (1 + \beta)^{-1} \approx \frac{1}{m_0} (1 - \beta) \quad (12)$$

Geometric interpretation: The factor 2 in $r_0 = 2Gm$ emerges from the relativistic field structure, exactly matching the Schwarzschild radius.

Modified Field Equation for Infinite Systems

For infinite, homogeneous fields, the field equation requires modification:

$$\nabla^2 m(x, t) = 4\pi G \rho_0 m(x, t) + \Lambda_T m(x, t) \quad (13)$$

where the consistency condition for homogeneous background gives:

$$\Lambda_T = -4\pi G \rho_0 \quad (14)$$

Dimensional verification: $[\Lambda_T] = [4\pi G \rho_0] = [1][E^{-2}][E^4] = [E^2] \checkmark$

This modification leads to the cosmic screening effect: $\xi_{\text{eff}} = \xi/2$.

.3 Lagrangian Formulation with Dimensional Consistency

Time Field Lagrangian Density

The fundamental Lagrangian density for the intrinsic time field is:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t) - V(T(x, t)) \right] \quad (15)$$

Dimensional verification:

- $[\sqrt{-g}] = [E^{-4}]$ (4D volume element)
- $[g^{\mu\nu}] = [E^2]$ (inverse metric)
- $[\partial_\mu T(x, t)] = [E][E^{-1}] = [1]$ (dimensionless gradient)
- $[g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t)] = [E^2][1][1] = [E^2]$
- $[V(T(x, t))] = [E^4]$ (potential energy density)
- Total: $[E^{-4}][E^2 + E^4] = [E^{-2}] + [E^0] \checkmark$

Modified Schrödinger Equation

The quantum mechanical evolution equation becomes:

$$iT(x, t) \frac{\partial}{\partial t} \Psi + i\Psi \left[\frac{\partial T(x, t)}{\partial t} + \vec{v} \cdot \nabla T(x, t) \right] = \hat{H} \Psi \quad (16)$$

Dimensional verification:

- $[iT(x, t)\partial_t\Psi] = [E^{-1}][E][\Psi] = [\Psi]$
- $[i\Psi\partial_t T(x, t)] = [\Psi][E^{-1}][E] = [\Psi]$
- $[\hat{H}\Psi] = [E][\Psi] = [\Psi] \checkmark$

Higgs Field Coupling

The Higgs field couples to the time field through:

$$\mathcal{L}_{\text{Higgs-T}} = |D_{\text{Higgs-T}}|^2 - V(T(x, t), \Phi) \quad (17)$$

where:

$$D_{\text{Higgs-T}} = T(x, t)(\partial_\mu + igA_\mu)\Phi + \Phi\partial_\mu T(x, t) \quad (18)$$

This establishes the fundamental connection:

$$T(x, t) = \frac{1}{y\langle\Phi\rangle} \quad (19)$$

.4 Matter Field Coupling Through Conformal Transformations

Conformal Coupling Principle

All matter fields couple to the time field through conformal transformations of the metric:

$$g_{\mu\nu} \rightarrow \Omega^2(T(x, t))g_{\mu\nu}, \quad \text{where} \quad \Omega(T(x, t)) = \frac{T_0}{T(x, t)} \quad (20)$$

Dimensional verification: $[\Omega(T(x, t))] = [T_0/T(x, t)] = [E^{-1}]/[E^{-1}] = [1]$ (dimensionless) $\checkmark$

Scalar Field Lagrangian

For scalar fields:

$$\mathcal{L}_\phi = \sqrt{-g}\Omega^4(T(x, t))\left(\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - \frac{1}{2}m^2\phi^2\right) \quad (21)$$

Dimensional verification:

- $[\Omega^4(T(x, t))] = [1]$ (dimensionless)
- $[g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi] = [E^2][E^2] = [E^4]$

- $[m^2\phi^2] = [E^2][E^2] = [E^4]$
- Total: $[E^{-4}][1][E^4] = [E^0]$ (dimensionless) ✓

Fermion Field Lagrangian

For fermion fields:

$$\mathcal{L}_\psi = \sqrt{-g}\Omega^4(T(x, t)) \left(i\bar{\psi}\gamma^\mu\partial_\mu\psi - m\bar{\psi}\psi \right) \quad (22)$$

Dimensional verification:

- $[i\bar{\psi}\gamma^\mu\partial_\mu\psi] = [E^{3/2}][1][E][E^{3/2}] = [E^4]$
- $[m\bar{\psi}\psi] = [E][E^{3/2}][E^{3/2}] = [E^4]$
- Total: $[E^{-4}][1][E^4] = [E^0]$ (dimensionless) ✓

.5 Connection to Higgs Physics and Parameter Derivation

The Universal Scale Parameter from Higgs Physics

The T0 model's fundamental scale parameter is uniquely determined through quantum field theory and Higgs physics. The complete calculation yields:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4} \quad (23)$$

where:

- $\lambda_h \approx 0.13$ (Higgs self-coupling, dimensionless)
- $v \approx 246$ GeV (Higgs VEV, dimension $[E]$)
- $m_h \approx 125$ GeV (Higgs mass, dimension $[E]$)

Complete dimensional verification:

$$[\xi] = \frac{[1][E^2]}{[1][E^2]} = \frac{[E^2]}{[E^2]} = [1] \quad (\text{dimensionless}) \quad (24)$$

Universal Scale Parameter

Key Insight: The parameter $\xi(m) = 2Gm/\ell_P$ scales with mass, revealing the **fundamental unity of geometry and mass**. At the Higgs mass scale, $\xi_0 \approx 1.33 \times 10^{-4}$ provides the natural reference value that characterizes the coupling strength between the time field and physical processes in the T0 model.

Connection to β_T Parameter

The relationship between the scale parameter and the time field coupling is established through:

$$\beta_T = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2 \xi} = 1 \quad (25)$$

This relationship, combined with the condition $\beta_T = 1$ in natural units, uniquely determines ξ and eliminates all free parameters from the theory.

Geometric Modifications for Different Field Regimes

The universal scale parameter ξ undergoes geometric modifications depending on the field configuration:

- **Localized fields:** $\xi = 1.33 \times 10^{-4}$ (full value)
- **Infinite homogeneous fields:** $\xi_{\text{eff}} = \xi/2 = 6.7 \times 10^{-5}$ (cosmic screening)

This factor of 1/2 reduction arises from the Λ_T term in the modified field equation for infinite systems and represents a fundamental geometric effect rather than an adjustable parameter.

.6 Complete Total Lagrangian Density

Full T0 Model Lagrangian

The complete Lagrangian density for the T0 model is:

$$\mathcal{L}_{\text{Total}} = \mathcal{L}_{\text{time}} + \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\phi} + \mathcal{L}_{\psi} + \mathcal{L}_{\text{Higgs-T}} \quad (26)$$

where each component is dimensionally consistent:

$$\mathcal{L}_{\text{time}} = \sqrt{-g} \left[\frac{1}{2} g^{\mu\nu} \partial_\mu T(x, t) \partial_\nu T(x, t) - V(T(x, t)) \right] \quad (27)$$

$$\mathcal{L}_{\text{gauge}} = \sqrt{-g} \left(-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \right) \quad (28)$$

$$\mathcal{L}_\phi = \sqrt{-g} \Omega^4(T(x, t)) \left(\frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m^2 \phi^2 \right) \quad (29)$$

$$\mathcal{L}_\psi = \sqrt{-g} \Omega^4(T(x, t)) \left(i \bar{\psi} \gamma^\mu \partial_\mu \psi - m \bar{\psi} \psi \right) \quad (30)$$

$$\mathcal{L}_{\text{Higgs-T}} = \sqrt{-g} |D_{\text{Higgs-T}}|^2 - V(T(x, t), \Phi) \quad (31)$$

Dimensional consistency: Each term has dimension $[E^0]$ (dimensionless), ensuring proper action formulation.

7 Cosmological Applications

Modified Gravitational Potential

The T0 model predicts a modified gravitational potential:

$$\Phi(r) = -\frac{GM}{r} + \kappa r \quad (32)$$

where κ depends on the field geometry:

- **Localized systems:** $\kappa = \alpha_\kappa H_0 \xi$
- **Cosmic systems:** $\kappa = H_0$ (Hubble constant)

Static Universe Interpretation

The T0 model explains cosmological observations without spatial expansion:

- **Redshift:** Energy loss to time field gradients
- **Cosmic microwave background:** Equilibrium radiation in static universe
- **Structure formation:** Gravitational instability with modified potential
- **Dark energy:** Emergent from Λ_T term in field equation

.8 Experimental Predictions and Tests

Distinctive T0 Signatures

The T0 model makes specific testable predictions using the universal scale parameter $\xi \approx 1.33 \times 10^{-4}$:

1. Wavelength-dependent redshift:

$$\frac{z(\lambda_2) - z(\lambda_1)}{z_0} = \ln \frac{\lambda_2}{\lambda_1} \quad (33)$$

2. QED corrections to anomalous magnetic moments:

$$a_\ell^{(T0)} = \frac{\alpha}{2\pi} \xi^2 I_{\text{loop}} \approx 2.3 \times 10^{-10} \quad (34)$$

3. Modified gravitational dynamics:

$$v^2(r) = \frac{GM}{r} + \kappa r^2 \quad (35)$$

4. Energy-dependent quantum effects:

$$\Delta t = \frac{\xi}{c} \left(\frac{1}{E_1} - \frac{1}{E_2} \right) \frac{2Gm}{r} \quad (36)$$

Precision Tests

The fixed-parameter nature allows stringent tests:

- **No free parameters:** All coefficients derived from $\xi \approx 1.33 \times 10^{-4}$
- **Cross-correlation:** Same parameters predict multiple phenomena
- **Universal predictions:** Same ξ value applies across all physical processes
- **Quantum-gravitational connection:** Tests of unified framework

Equation	Left Side	Right Side	Status
Time field definition	$[T] = [E^{-1}]$	$[1/\max(m, \omega)] = [E^{-1}]$	✓
Field equation	$[\nabla^2 m] = [E^3]$	$[4\pi G \rho m] = [E^3]$	✓
β parameter	$[\beta] = [1]$	$[2Gm/r] = [1]$	✓
ξ parameter (Higgs)	$[\xi] = [1]$	$[\lambda_h^2 v^2 / (16\pi^3 m_h^2)] = [1]$	✓
β_T relationship	$[\beta_T] = [1]$	$[\lambda_h^2 v^2 / (16\pi^3 m_h^2 \xi)] = [1]$	✓
Energy loss rate	$[dE/dr] = [E^2]$	$[g_T \omega^2 2G/r^2] = [E^2]$	✓
Modified potential	$[\Phi] = [E]$	$[GM/r + \kappa r] = [E]$	✓
Lagrangian density	$[\mathcal{L}] = [E^0]$	$[\sqrt{-g} \times \text{density}] = [E^0]$	✓
QED correction	$[a_t^{(T0)}] = [1]$	$[\alpha \xi^2 / 2\pi] = [1]$	✓

Table 1: Complete dimensional consistency verification for T0 model equations

.9 Dimensional Consistency Verification

Complete Verification Table

.10 Connection to Quantum Field Theory

Modified Dirac Equation

The Dirac equation in the T0 framework becomes:

$$[i\gamma^\mu(\partial_\mu + \Gamma_\mu^{(T)}) - m(x, t)]\psi = 0 \quad (37)$$

where the time field connection is:

$$\Gamma_\mu^{(T)} = \frac{1}{T(x, t)} \partial_\mu T(x, t) = -\frac{\partial_\mu m}{m^2} \quad (38)$$

QED Corrections with Universal Scale

The time field introduces corrections to QED calculations using the universal scale parameter:

$$a_e^{(T0)} = \frac{\alpha}{2\pi} \cdot \xi^2 \cdot I_{\text{loop}} = \frac{1}{2\pi} \cdot (1.33 \times 10^{-4})^2 \cdot \frac{1}{12} \approx 2.34 \times 10^{-10} \quad (39)$$

This prediction applies universally to all leptons, reflecting the fundamental nature of the scale parameter.